

Lecture 3 — Elimination and the LU Decomposition

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After this lecture you should be able to

- Carry out Gaussian elimination on a small system and finish it by back substitution.
- Write a single row operation as multiplication by a lower triangular matrix, and read off $A = LU$.
- Eliminate on T and on K and write down the resulting L and U .
- Explain why a symmetric matrix comes out of elimination as LDL^T .
- Read elimination on a resistor ladder as node elimination, a pivot as the conductance that is left, and one LU as the factorisation reused at every step of a transient.

From building matrices to solving with them

Last lecture produced four matrices. None of them is any use until we can answer $Ax = b$. Today's answer is the one every piece of numerical software actually uses, and the point of the lecture is that it is not really an algorithm — it is a *factorisation*. Once you see elimination as $A = LU$, the arithmetic you did by hand becomes a reusable object.

1 Gaussian elimination: two views of one calculation

The standard technique for a general linear system is Gaussian elimination. Here it is twice: once as the variable elimination you already know, once as row reduction of a matrix.

Variable elimination

$$\begin{aligned}x + 3y &= -3 \\ -2x + y &= -8\end{aligned}$$

↓ cancel out the x term in the second equation

$$\begin{aligned}x + 3y &= -3 \\ 7y &= -14\end{aligned}$$

Matrix row reduction

$$\begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -3 \\ -8 \end{bmatrix}$$

↓ make the matrix **upper triangular**

$$\begin{bmatrix} 1 & 3 \\ 0 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -3 \\ -14 \end{bmatrix}$$

Either way, the last equation now involves one unknown, so we solve upwards — **back substitution**:

$$7y = -14 \implies y = -2, \quad x + 3(-2) = -3 \implies x = 3.$$

The general idea

Given a system $Ax = b$, do row operations on A (and on b) to obtain an **upper triangular** system, which can then be solved by back substitution. The two columns above are the same calculation; the right-hand one just stops writing the letters x and y down.

2 The LU decomposition: elimination as matrix multiplication

The LU decomposition is the matrix representation of Gaussian elimination. To get there, notice that the row operation we just performed is itself a matrix multiplication. Start from

$$\underbrace{\begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_b = \underbrace{\begin{bmatrix} -3 \\ -8 \end{bmatrix}}_b$$

and multiply both sides on the left by the matrix that adds twice the first row to the second:

$$\underbrace{\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}}_{L^{-1}} \underbrace{\begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{L^{-1}b} = \underbrace{\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}}_{L^{-1}} \underbrace{\begin{bmatrix} -3 \\ -8 \end{bmatrix}}_{L^{-1}b} \implies \underbrace{\begin{bmatrix} 1 & 3 \\ 0 & 7 \end{bmatrix}}_U \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{L^{-1}b} = \underbrace{\begin{bmatrix} -3 \\ -14 \end{bmatrix}}_{L^{-1}b}.$$

So the whole elimination is the single statement

$$L^{-1}A = U \iff \boxed{A = LU}$$

Note that the inverse of a lower triangular matrix is still lower triangular, so L is lower triangular too. Here inverting the step just flips the sign of the multiplier, giving $L = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$, and indeed

$$\begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ -2 & -6+7 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix} = A.$$

What the factorisation buys you

L is a record of the elimination steps and U is the triangular system they produced. Solving $Ax = b$ becomes two triangular solves: $Lc = b$ forwards, then $Ux = c$ backwards. The expensive part — the elimination — depended only on A , so a second right-hand side is nearly free.

More than one step

Larger matrices need several row operations. Applying them one after another to $Ax = b$ gives

$$\underbrace{(L_n^{-1} \cdots L_1^{-1})}_{\text{gives an upper triangular } U} Ax = (L_n^{-1} \cdots L_1^{-1})b,$$

so that $L_n^{-1} \cdots L_1^{-1}A = U$, and therefore

$$A = (L_1 \cdots L_n)U = LU.$$

The individual matrices L_j and L_j^{-1} for any single row operation are simple to write down and to invert. For example,

$$L_j^{-1} = \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ \vdots & \ell_j & \ddots & \\ 0 & 0 & \cdots & 1 \end{bmatrix}, \quad L_j = \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ \vdots & -\ell_j & \ddots & \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

— just a change in sign of the factor.

Do not assemble L by guessing

The inverse of a product of several L_j is not necessarily as simple as putting the various $-\ell_j$ together in their various positions. It depends on the order in which the steps were taken. Work a 3×3 case out by hand before you trust the pattern — this is on the problem set.

3 Worked example: free–fixed centred differences, T

Take the free–fixed second-difference matrix from last lecture, here at size 4:

$$\begin{aligned}
 T &= \begin{bmatrix} 1 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & -1 & & \\ 0 & 1 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \quad \ell_{21} = 1 \\
 &\longrightarrow \begin{bmatrix} 1 & -1 & & \\ 0 & 1 & -1 & \\ & 0 & 1 & -1 \\ & & -1 & 2 \end{bmatrix} \quad \ell_{32} = 1 \quad \longrightarrow \begin{bmatrix} 1 & -1 & & \\ 0 & 1 & -1 & \\ & 0 & 1 & -1 \\ & & 0 & 1 \end{bmatrix} \quad \ell_{43} = 1.
 \end{aligned}$$

Every pivot came out equal to 1 and every multiplier equal to 1, so

$$L = \begin{bmatrix} 1 & & & \\ -1 & 1 & & \\ & -1 & 1 & \\ & & -1 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & -1 & & \\ & 1 & -1 & \\ & & 1 & -1 \\ & & & 1 \end{bmatrix}, \quad T = LU.$$

Look at those two factors. U is exactly $L^\top$, and both are bidiagonal: a single first difference. The second-difference matrix has come apart into two first-difference matrices — which is precisely how we built it in Lecture 1, now recovered by elimination rather than by construction.

4 Worked example: fixed–fixed centred differences, K

Now the fixed–fixed matrix, where the corner entries are 2 instead of 1 and the arithmetic is less tidy:

$$\begin{aligned}
 K &= \begin{bmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & -1 & & \\ 0 & \frac{3}{2} & -1 & \\ & -1 & 2 & -1 \\ & & -1 & 2 \end{bmatrix} \quad \ell_{21} = \frac{1}{2} \\
 &\longrightarrow \begin{bmatrix} 2 & -1 & & \\ 0 & \frac{3}{2} & -1 & \\ & 0 & \frac{4}{3} & -1 \\ & & -1 & 2 \end{bmatrix} \quad \ell_{32} = \frac{2}{3} \quad \longrightarrow \begin{bmatrix} 2 & -1 & & \\ 0 & \frac{3}{2} & -1 & \\ & 0 & \frac{4}{3} & -1 \\ & & 0 & \frac{5}{4} \end{bmatrix} \quad \ell_{43} = \frac{3}{4},
 \end{aligned}$$

the last of which is U . Collecting the multipliers with their signs flipped,

$$L = \begin{bmatrix} 1 & & & \\ -\frac{1}{2} & 1 & & \\ & -\frac{2}{3} & 1 & \\ & & -\frac{3}{4} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & -1 & & \\ & \frac{3}{2} & -1 & \\ & & \frac{4}{3} & -1 \\ & & & \frac{5}{4} \end{bmatrix}.$$

The pivots are not random

The pivots came out $2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}$ — the pattern $\frac{k+1}{k}$. What would the next one be if we had eliminated on the 5×5 version? Structured matrices give structured factorisations, and that is exactly why it is worth knowing that a matrix is $\mathbf{K}$ rather than just “some tridiagonal matrix”.

5 Symmetry survives: $\mathbf{K} = \mathbf{LDL}^\top$

The $\mathbf{L}$ and $\mathbf{U}$ above do not look like transposes of each other, but they nearly are: $\mathbf{U}$ is $\mathbf{L}^\top$ with the rows rescaled. So extract a factor from each row of $\mathbf{U}$ to make the diagonal entries equal to 1:

$$\mathbf{U} = \underbrace{\begin{bmatrix} 2 & & & & \\ & \frac{3}{2} & & & \\ & & \frac{4}{3} & & \\ & & & \frac{5}{4} & \\ & & & & 1 \end{bmatrix}}_{\mathbf{D}} \underbrace{\begin{bmatrix} 1 & -\frac{1}{2} & & & \\ & 1 & -\frac{2}{3} & & \\ & & 1 & -\frac{3}{4} & \\ & & & 1 & \\ & & & & 1 \end{bmatrix}}_{\text{this is } \mathbf{L}^\top}.$$

Therefore

$$\boxed{\mathbf{K} = \mathbf{LDL}^\top}$$

The $\mathbf{LDL}^\top$ form is not surprising, given that we started with a symmetric matrix $\mathbf{K}$. And $\mathbf{LDL}^\top$ is *manifestly* symmetric — you do not have to compute anything to see it:

$$(\mathbf{LDL}^\top)^\top = (\mathbf{L}^\top)^\top \mathbf{D}^\top \mathbf{L}^\top = \mathbf{LDL}^\top,$$

using only that $\mathbf{D}$ is diagonal, hence its own transpose.

Symmetry in, symmetry out

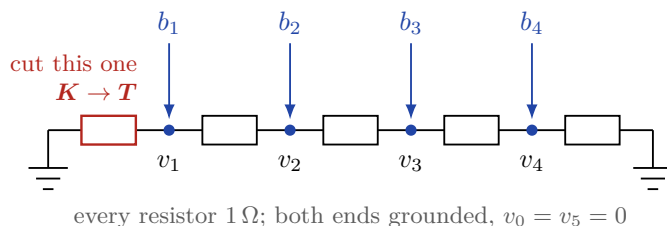
Elimination on a symmetric matrix does not destroy its symmetry; it relocates it. All of the asymmetry of $\mathbf{LU}$ was bookkeeping — the scaling of the rows — and pulling that scaling out into $\mathbf{D}$ leaves the same triangular factor on both sides. The information that used to be spread over the whole matrix now sits in one triangle and one diagonal.

Watch the diagonal

All four pivots of $\mathbf{K}$ came out positive. That is not an accident either, and it is not just aesthetic: when we get to positive definite matrices in Lecture 8, the signs of the entries of $\mathbf{D}$ will be the test we run.

6 Physical example: $\mathbf{K}$ and $\mathbf{T}$ as a resistor ladder

The same two matrices come out of a circuit, and the circuit explains the pivots. Put four nodes in a line, join each neighbouring pair by a $1\ \Omega$ resistor, ground both ends, and inject a current b_k into node k . The unknowns are the node voltages v_k .



Kirchhoff's current law at node k says that the current injected there leaves through the two resistors touching it, and Ohm's law turns each of those currents into a voltage difference:

$$\frac{v_k - v_{k-1}}{1\ \Omega} + \frac{v_k - v_{k+1}}{1\ \Omega} = b_k \quad \Longleftrightarrow \quad -v_{k-1} + 2v_k - v_{k+1} = b_k.$$

That is the interior row of $\mathbf{K}$, with $v_0 = v_5 = 0$ at the grounded ends. Cut the resistor at the left end instead and node 1 has only one resistor attached, so its row loses a term and reads $v_1 - v_2 = b_1$ — the 1 in the corner of $\mathbf{T}$. This is Lecture 2's key idea in electrical dress: a grounded node is a held end, an open node is a free end, and the whole difference is one number in one corner.

A pivot is the conductance you have left

Eliminating node 1 *is* deleting node 1 from the circuit: its equation is used to write v_1 in terms of v_2 , and what remains is the network the other nodes still see. So each pivot is the total conductance at the node being eliminated, looking into whatever is still attached to it.

Walk that down the ladder of $\mathbf{K}$. Node 1 sees 1 S to ground and 1 S to node 2: pivot 2. With node 1 absorbed, node 2 looks left through *two* unit resistors in series, worth $\frac{1}{2}$ S, and still sees 1 S on its right: pivot $\frac{3}{2}$. After $k - 1$ eliminations the path to ground on the left is k resistors in series, so

$$\text{pivot}_k = \underbrace{1}_{\text{right}} + \underbrace{\frac{1}{k}}_{\text{left}} = \frac{k+1}{k},$$

which is the pattern of the previous aside, now with a reason — and the fifth pivot must be $\frac{6}{5}$. For $\mathbf{T}$ the left end is open, so the chain behind node k is a dead end carrying no current and contributing no conductance; every pivot is the single unit conductance on the right, 1, exactly as the arithmetic gave.

The triangular solves are physical too. For $\mathbf{T} = \mathbf{L}\mathbf{U}$ with $\mathbf{U} = \mathbf{L}^\top$, the forward solve $\mathbf{L}\mathbf{c} = \mathbf{b}$ reads $c_k - c_{k-1} = b_k$, so $c_k = b_1 + \dots + b_k$: with the far end open, every current injected to the left of resistor k has nowhere to go but through it, and c_k is that current. The back solve $\mathbf{L}^\top \mathbf{v} = \mathbf{c}$ then reads $v_k - v_{k+1} = c_k$, which is Ohm's law across that resistor. The factors are the ladder's difference matrix $\mathbf{A}$ and its transpose, $\mathbf{T} = \mathbf{A}^\top \mathbf{A}$, and elimination has split the problem into conservation first and constitutive law second — the split that will organise Lecture 12 onwards.

7 Physical example: an RC ladder, one factorisation for every time step

Hang a capacitor C from each node of the same ladder down to ground, keep both ends of the ladder grounded, and let the injected currents vary in time. Kirchhoff's law at node k now carries a charging term, and in vector form

$$C \frac{d\mathbf{v}}{dt} + \frac{1}{R} \mathbf{K} \mathbf{v} = \mathbf{b}(t).$$

Step it forward implicitly — evaluate the resistive term at the new time, which is backward Euler, properly in Lecture 14:

$$C \frac{\mathbf{v}^{n+1} - \mathbf{v}^n}{\Delta t} + \frac{1}{R} \mathbf{K} \mathbf{v}^{n+1} = \mathbf{b}^{n+1} \implies \boxed{(\mathbf{I} + \alpha \mathbf{K}) \mathbf{v}^{n+1} = \mathbf{v}^n + \frac{\Delta t}{C} \mathbf{b}^{n+1}}, \quad \alpha = \frac{\Delta t}{RC}.$$

Every time step is a solve with the same $\mathbf{M} = \mathbf{I} + \alpha \mathbf{K}$: tridiagonal, symmetric, and with a heavier diagonal than $\mathbf{K}$ itself. Taking $\alpha = 1$, a step of one time constant RC ,

$$\mathbf{M} = \begin{bmatrix} 3 & -1 & & \\ -1 & 3 & -1 & \\ & -1 & 3 & -1 \\ & & -1 & 3 \end{bmatrix}, \quad \mathbf{M} = \mathbf{L} \mathbf{D} \mathbf{L}^\top, \quad \mathbf{D} = \text{diag}\left(3, \frac{8}{3}, \frac{21}{8}, \frac{55}{21}\right),$$

with multipliers $\ell_{21} = \frac{1}{3}$, $\ell_{32} = \frac{3}{8}$, $\ell_{43} = \frac{8}{21}$ in the same convention as above. Symmetric matrix in, $\mathbf{L} \mathbf{D} \mathbf{L}^\top$ out, all four pivots positive — one triangle and one diagonal is the whole factorisation.

Now the point. $\mathbf{M}$ is built from the circuit and the step size; the right-hand side $\mathbf{v}^n + (\Delta t/C) \mathbf{b}^{n+1}$ is built from the present state and the input signal. Only the second changes as the simulation runs. So factor $\mathbf{M}$ *once*, before the loop, and spend each of the thousand time steps on two triangular solves — which is the “a second right-hand side is nearly free” idea from the beginning of the lecture, now paying for itself a thousand times over.

Never form $\mathbf{M}^{-1}$, and watch Δt

$\mathbf{M}^{-1}$ is a full matrix, while $\mathbf{L}$ and $\mathbf{U}$ keep the two nonzero diagonals of $\mathbf{M}$: forming the inverse throws away the sparsity you started with. And α sits *inside* $\mathbf{M}$, so changing Δt means a new matrix and a new factorisation — which is exactly why implicit solvers hold the step fixed for long stretches. Here the factorisation and the solve are both $\mathcal{O}(n)$ and the saving is a constant factor; on the two-dimensional grid of Lecture 17 the factorisation costs far more than the solve, and re-using it is the difference between a simulation that runs and one that does not.

Where this is going

Elimination answers “solve $\mathbf{A}\mathbf{x} = \mathbf{b}$ ” mechanically, without ever asking what the columns of $\mathbf{A}$ are doing. Next time we ask: reading $\mathbf{A}\mathbf{x} = \mathbf{b}$ as a request to build $\mathbf{b}$ out of the columns of $\mathbf{A}$, when is that possible, when is it unique, and what changes if we choose a different set of building blocks?

Reading

Strang, *Computational Science and Engineering*, §1.3.